MIL-H-43606B(GL)
10 June 1977
SUPERSEDING
MIL-H-43606A(GL)
29 June 1971

MILITARY SPECIFICATION

HAND CLEANER, WATERLESS

This specification is approved for use by Natick R&D Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for one type of a waterless hand cleaner (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-P-94	-	Paper, Test; pH Indicator.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.

STANDARDS

FEDERAL

FED-STD-595	-	Color.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8520

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

16 CFR 1500 - Federal Hazardous Substances Act.
21 CFR - Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 287 - API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method)
D 156 - Saybolt Color of Petroleum Products (Saybolt Chromometer Method)
D 611 - Aniline Point and Mixed Aniline Point of Petroleum Products and Hydrocarbon Solvent
D 1266 - Sulphur in Petroleum Products (Lamp Method)
D 86 - Distillation of Petroleum Products

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. All ingredients used in the manufacture of the hand cleaner shall be odorless, either white or colorless, of a type and grade applicable for use on the skin, and shall conform to the requirements of the Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder (see 6.3). Kerosene, chlorinated solvents, ammonia, free caustic, and abrasives are prohibited.

* 3.2 Composition. The composition of the hand cleaner shall conform to the requirements of table I. See 6.4 for mixing directions.

TABLE I. Composition

Ingredient	Percent by weight
Isoparaffinic hydrocarbon (technical)	31.0 ± 1.0
Light mineral oil (white, domestic, saybolt viscosity 125/135)	15.7 ± 0.5
Stearic acid (USP triple distilled)	8.0 ± 0.2
Mixed Isopropanolamine (technical)	4.0 ± 0.1
Distilled or deionized water	36.0 ± 1.0
Glycerine (USP)	3.0 ± 0.1
Sodium tetradecyl sulfate (technical)	1.0 ± 0.05
Polyvinylpyrrolidone (technical)	0.8 ± 0.05

* 3.2.1 Hydrocarbon solvent. The hydrocarbon solvent shall be an isoparaffinic hydrocarbon conforming to the following specific requirements when tested as specified in 4.3.4:

	Min.	Max.
Specific gravity API 60°/60°F. (15°/15°C)	54.0	-
Color, Saybolt	30	-
Aniline point °F (°C)	180 (82)	-
Sulphur ppm	-	10
Distillation range °F (°C)	340 (171)	375 (191)

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3.3 Color and texture. The hand cleaner shall be equal to or whiter than Color Chip No. 17886 of FED-STD-595, and shall be a smooth, homogeneous cream.

3.4 Stability (freeze-thaw and heat tests). The hand cleaner shall show no separation, and shall develop no color change or rancidity when tested as specified in 4.3.4.

3.5 (pH). The hand cleaner shall have a pH of 5.5 to 8.0 when tested as specified in 4.3.4.

3.6 Odor. The hand cleaner shall leave no objectionable odor on the hands when tested as specified in 4.3.4.

3.7 Soil removal. The hand cleaner shall effectively remove soil and shall have a rating of "good" or "excellent" when tested as specified in 4.3.4.

3.8 Grit. The hand cleaner shall be free of grit when tested as specified in 4.3.4.

3.9 Labeling. Labeling shall conform to the regulations under the Federal Hazardous Substances Act. In addition to the regular marking, each unit container shall be durably and legibly marked with the following special marking:

Instructions for Use

1. Use sparingly for light soil. Place small amount of cleaner on the hands. Rub briskly until cleaner liquefies and dirt, grime, or smudges are loosened.

2. Wipe dry with paper towels or other available materials.

WARNING

Keep out of reach of children.
Do not take internally.
Do not use if hands or arms are susceptible to skin rash.

* 3.10 Workmanship. The hand cleaner shall be clean and shall not contain dirt or foreign matter. In addition, the container shall not be dented, abraded, dirty, split or punctured and there shall be no seepage of contents.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, a lot shall consist of all hand cleaner manufactured as one batch, packaged alike, and submitted for inspection at one time.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended or qualified in this specification or applicable purchase documents. The contractor shall submit to the contracting officer or his authorized representative a certificate of compliance covering the requirements in 3.2 except the hydrocarbon solvent specified in 3.2.1 which shall be tested as specified in 4.3.4.

4.3 Inspection of the end item.

* 4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs and at the inspection levels and the acceptable quality levels (AQLs) set forth in 4.3.3. Each lot of end items shall be examined for visual and net contents defects. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers for the examinations in 4.3.1.1 and 4.3.1.2.

4.3.1.1 Examinations for visual defects. The sample unit for this examination shall be one filled unit container of hand cleaner.

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Examine	Defect
Unit container:	
Construction	Not as specified (see 5.1).
Closure	Not as specified (see 5.1); imperfect.
Workmanship	Dented, abraded, dirty, split, punctured; any seepage of contents.
Markings	Missing, incomplete, incorrect, not in accordance with contract requirements. Does not contain labeling information specified in 3.9.
Contents:	
Color and texture	Color not as specified. Contains lumps; not smooth, not homogeneous.
Workmanship	Not clean, presence of dirt or foreign matter.

4.3.1.2 Examination of filled containers for net contents. The sample unit for this examination shall be one filled unit container of hand cleaner. The lot shall be unacceptable if the average net contents per container, for all sample units examined, is less than required.

* 4.3.2 Packaging inspection. An examination shall be made to determine that preservation-packaging, packing, and marking comply with section 5 requirements. Defects shall be as listed below. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully prepared for delivery. The lot shall be the number of shipping containers in the end item inspection lot. The inspection levels shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; incomplete; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged or not as specified.

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Examine	Defect
Workmanship	Inadequate application of components, such as incomplete closure of container flap, improper taping, loose strapping, inadequate stapling. Bulged or distorted container.
Contents	Number of unit packages per shipping container is less than specified.

4.3.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A

4.3.4 Testing of the end item. The methods of testing listed in table II shall be followed for each lot. The lot size shall be expressed in the number of unit containers. Each lot shall consist of a single composition of product, manufactured under essentially the same conditions, and at essentially the same time. The sample unit for testing shall be one filled unit container. The sample size shall be as indicated in the table below. The lot shall be unacceptable if any test sample fails to meet the test requirement specified.

Lot size (number of unit containers)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	2/ Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver	Per sample unit	Pass or Fail	Numerically to Nearest		
Composition	3.2	1/							
Stability (freeze-thaw, and heat tests)	3.4	4.4.1 and 4.4.2	X		1	X			
(pH)	3.5	4.4.3	X		2	X 3/			
Odor	3.6	4.4.4	X		1	X			
Soil removal	3.7	4.4.5	X		1	X			
Hydrocarbon solvent	3.2.1	4.4.6	X		1	X			
Grit	3.8	4.4.7	X		1	X			

1/ The contractor shall submit to the contracting officer, or his authorized representative, production data certifying the compliance of each prepared batch with composition requirements.

2/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.

3/ Both tests must pass this requirement.

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4.4 Test procedures.

* 4.4.1 Freeze-thaw test. A 25.0 ± 0.5 g sample of hand cleaner, in a 1 ounce (28 g) capped bottle, shall be exposed to a temperature of -40° ± 2°F (-40° ± 1°C) for 18 hours. The sample shall then be allowed to remain at room temperature until thawed. This test shall be repeated to five complete cycles, unless separation, rancidity, or a sharp color change occurs.

* 4.4.2 Heat test. A 25.0 ± 0.5 g sample of hand cleaner, in a 1 ounce (28 g) capped bottle shall be exposed to a temperature of 155-160°F (68-71°C) for four hours. The sample shall then be allowed to cool to room temperature. This test shall be repeated for five complete cycles, unless separation, rancidity, or a sharp color change occurs.

4.4.3 (pH). A small quantity of hand cleaner shall be applied to both sides of a strip of pH indicator test paper conforming to pH range e, class A or B of O-P-94, and the color in the paper allowed to develop. This color is compared with the color of the color comparison chart. The indicated pH must be in the range specified in 3.5.

4.4.4 Odor. A small quantity of the hand cleaner shall be examined to detect any objectionable odor. Approximately 1.0 g of the hand cleaner shall be rubbed on one hand and the excess removed with a paper towel. There shall be only a faint remaining pleasant odor on the skin.

4.4.5 Soil removal.

* 4.4.5.1 Soiling mixture. The test soil shall be prepared by mixing thoroughly the following ingredients (see 6.5):

1 quart (945 ml) SAE #20 non detergent lubricating oil
1/2 pound (226 g) lime grease
4 ounces (113 g) carbon lamp black
6 ounces (170 g) charcoal black

4.4.5.2 Soil removal test. Soil removal shall be determined as follows:

Step 1. Wash both hands with soap and water, dry, and wait 15 minutes.

Step 2. Apply approximately 0.1 g of the soiling mixture to the back of the left hand and knuckles with the right forefinger. Rub the soiling mixture into the skin by stroking lengthwise, crosswise, and in a circular motion for 1 minute.

Step 3. Remove any excess soiling mixture by wiping with a paper towel.

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Step 4. Rub 1.0 g of the hand cleaner under test into the soiled skin area for 1 minute. Wipe the excess cleaner and soil from the hand with a paper towel.

Step 5. Repeat the operation described in step 4 and compare the treated hand with the unsoiled right hand. Rate the removal as excellent, good, fair or poor as follows:

Excellent - soiling mixture is completely removed with no trace of skin discoloration.

Good - substantially all the soiling mixture is removed with soil remaining in the creases of the knuckles.

Fair - soil remains in both the pores and knuckle creases.

Poor - soil remains in pores, knuckle creases, and on skin surface.

* 4.4.6 Hydrocarbon solvent. The specific properties of the isoparaffinic hydrocarbon shall be determined in accordance with the following ASTM Test Methods:

Property	Test methods
Specific gravity	ASTM D 287
Color, Syabolt	ASTM D 156
Aniline point °F (°C)	ASTM D 611
Sulphur ppm	ASTM D 1266
Distillation range °F (°C)	ASTM D 86

4.4.7 Grit. Spread evenly approximately 1 gram of the hand cleaner on a clean microscope slide. Place a second clean microscope slide over the hand cleaner and press together with a rotary motion. There shall be no evidence of grit and the microscope slides shall not show any sign of being scratched after the cleaner is removed.

5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound (453 g) of hand cleaner shall be packaged in metal cans 404 x 308 round, conforming to type V, class 2 of PPP-C-96, except that terne plate, blackplate or 0.25 electrolytic tinplate shall not be used. The exterior coating of the cans shall conform to coating plan A of PPP-C-96. The

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interior coating of the cans shall conform to the protective coating requirement specified in PPP-C-96. The seaming compound shall not affect nor be affected by the cream hand cleaner or any of the liquid components of the hand cleaner. Alternatively to compound-lined double seamed ends and sides, the seams may be soldered.

* 5.1.2 Level C. Hand cleaner shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A packing. Hand cleaner, preserved-packaged as specified in 5.1, shall be packed in accordance with level A packing requirements for type V, friction plug cans in the appendix of PPP-C-96.

* 5.2.2 Level B packing. Hand cleaner, preserved-packaged as specified in 5.1, shall be packed in accordance with level B requirements for type V friction plug cans in the appendix of PPP-C-96.

* 5.2.3 Level C packing. Hand cleaner, preserved packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hand cleaner covered by this specification is intended for use by Civil Defense Agencies in survival kits. The hand cleaner is suitable for storage at extreme high and low ambient temperatures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

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* 6.3 Federal regulation. Matters pertaining to compliance of the hand cleaner under Federal Food, Drug, and Cosmetic Act should be referred to the Department of Health, Education, and Welfare, Food and Drug Administration, Rockville, MD 20852.

6.4 Mixing directions. The following mixing directions were followed:

Melt the stearic acid at 69-70°C. When melted, slowly add light mineral oil and isoparaffinic hydrocarbon. Then mix isopropanolamine, distilled water, glycerine, sodium tetradecyl sulfate, and polyvinylpyrrolidone and slowly add this mixture to the oil phase, with continuous stirring. After all ingredients have been combined, stir the emulsion until the ingredients are well blended. If stirred too vigorously or for a long period of time, air bubbles will develop and make it difficult or impossible to package one pound (453 g) of hand cleaner in a one pound (453 g) can.

* 6.5 Ingredients. Test soil ingredients were obtained from the following sources:

(a) SAE #20 Nondetergent lubricating oil - Ashland Oil Co., Inc. P. O. Box 391, Ashland, KY 41101.

(b) Lime grease - Atlantic Lubricant #2 Atlantic Richfield Company, 260 So. Broad Street, Philadelphia, PA 19101.

(c) Carbon lamp black - Lampblack #12, Cities Service Co., Columbian Division, 3200 West Market Street, Akron, OH 44313.

(d) Charcoal black - Forest Products Charcoal Co., 484 North Hollywood Street, Memphis, TN 38112.

* 6.6 Changes from previous issue. The outside margins of this specification are marked with an asterisk (*) to indicate where changes (additions, corrections, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE
HAND CLEANER, WATERLESS MIL-H-43606B(GL)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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